Lesson 3.9 — Fundamental Theorem of Algebra

Big idea: Every polynomial of degree n with complex coefficients has exactly n zeros, counted with multiplicity, in the complex numbers. No more, no fewer.

Quick review

FTA: A degree-n polynomial has n complex zeros (counting multiplicity).

Real polys: non-real zeros come in conjugate pairs ($a + bi$ and $a – bi$).

Counting consequence: a degree-3 real polynomial has 3 zeros total. Either all 3 real, or 1 real + a conjugate pair.

Linear factor form: $P(x) = a(x – r_1)(x – r_2)\ldots(x – r_n)$ (over ■).

Worked example

Worked example. Find all zeros of $P(x) = x^3 – x^2 + 4x – 4$.

Test $x = 1$: $1 – 1 + 4 – 4 = 0$ ✓. Synthetic → $x^2 + 4$. Set $= 0 \Rightarrow x = \pm 2i$.
Three zeros total: 1, 2i, –2i (1 real + a conjugate pair).

Warm-ups

1. How many complex zeros does $P(x) = 5x^4 – 3x^2 + 1$ have (with multiplicity)?
2. If a real polynomial has zeros 2 and $3 + i$, what other zero must it have?
3. A degree-3 real polynomial — how many real zeros must it have at least? Why?

Practice

1. Find all zeros of $x^3 + 2x^2 + x + 2$. (Hint: factor by grouping.)
2. Find all zeros of $x^4 – 16$. (Use difference of squares twice.)
3. Write a polynomial of lowest degree with real coefficients whose zeros include 1 and $2 – 3i$.

Challenge

1. Find all zeros of $P(x) = x^4 – 2x^3 + 6x^2 – 8x + 8$, given that $x = 1 + i$ is one zero.
2. Explain why a degree-5 real polynomial must have at least one real zero.

Answer key — Lesson 3.9

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. Four (degree 4).

2. $3 - i$ (the conjugate).

3. At least one real zero: end behavior of an odd-degree real polynomial goes from $-\infty$ to $+\infty$ (or reverse), so by IVT it must cross 0.

Practice

1. $x^2(x + 2) + 1(x + 2) = (x^2 + 1)(x + 2)$. Zeros: $x = -2$, $x = \pm i$.

2. $(x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$. Zeros: $2, -2, 2i, -2i$.

3. Need 1, $2 - 3i$, and $2 + 3i$. $P(x) = (x - 1)((x - 2)^2 + 9) = (x - 1)(x^2 - 4x + 13)$.

Challenge

1. Conjugate root theorem: $1 - i$ is also a zero. $(x - (1+i))(x - (1-i)) = x^2 - 2x + 2$. Divide P by $x^2 - 2x + 2 \rightarrow x^2 + 4$. Zeros of $x^2 + 4$ are $\pm 2i$. Four zeros: $1 \pm i$, $\pm 2i$.

2. End behavior of any odd-degree polynomial sends one end to $+\infty$ and the other to $-\infty$. By the Intermediate Value Theorem, the continuous polynomial must cross $y = 0$ at least once.